MIL-D-82659(OS)
31 January 1977
Superseding
(See Section 6)

MILITARY SPECIFICATION

DIETHYLENE GLYCOL DIBUTYL ETHER

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade diethylene glycol dibutyl ether referred to herein as "The material".

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipping and Storage
MIL-STD-1218	ACS Chemicals

PUBLICATIONS

Naval Sea Systems Command (Code Ident 10001)

OD 18893	Water Content, Determination of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to. Commanding Officer, Naval Ordnance Station, Standardization Division (611), Indian Head, MD 20640 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein Unless otherwise indicated, the issue in effect on the date of the invitation for bids or the request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 100-199 Transportation

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S Government Printing Office, Washington, DC 20402. Orders should specify "49 CFR 100-199 (latest revision)".)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn Tariff Order Section, 1616 P Street, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606)

(Technical society and tehnical association specifications and standards are generally available for reference from libraries They are distributed among technical groups and using Federal agencies.)

3 REQUIREMENTS

3 1 Material The material shall be diethylene glycol dibutyl ether in amber liquid form.

3.2 Chemical and physical properties The chemical and physical properties shall be in accordance with TABLE I

3 3 Workmanship The material shall be uniform, free from contaminants, foreign material or any other defect that would prevent its use for the purpose intended.

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TABLE I. Chemical and Physical Properties.

Property	Values	
	minimum	maximum
pH	6.0	7.5
Assay, volume %	95.0	-
Water content, wt%	-	0.30
Color	-	4
Specific Gravity, 25/25°C	0.960	0.980

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a	Temperature	Room ambient 18 to 35°C (65 to 95°F)
b	Altitude	Normal ground
c	Vibration	None
d	Humidity	Room ambient to 95 percent relative, maximum

4.3 Sampling.

4.3.1 Lot. Unless otherwise specified in the contract (see 6.2), a lot shall consist of all material manufactured in one continuous production run or in one batch, under essentially identical conditions, from the same raw materials, and to be offered for acceptance at one time. Several batches, manufactured from the same raw materials, may be blended to form a uniform larger batch which shall then constitute a lot for inspection purposes.

4.3.2 Sampling. Sampling for quality conformance inspection shall be in accordance with inspection Level I of MIL-STD-105. The sample unit shall be one unit package or container of material. Each sample shall consist of sufficient material to perform the quality conformance tests as specified in 4.4.

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4.4 Quality conformance inspection. Each sample obtained in accordance with 4.3.2 shall be subjected to the tests and examination of 4.5. The acceptable quality level (AQL) shall be 2.5% defective. When specified in the contract (see 6.2), the contractor shall furnish test reports showing quantitative results for all quality conformance tests specified for each lot of material.

4.5 Tests. Unless otherwise specified herein, all chemicals shall be ACS grade in accordance with MIL-STD-1218.

4.5.1 Visual examination. All samples shall be visually examined to verify conformance to the workmanship requirements.

4.5.2 pH. pH shall be determined in accordance with the following.

a. Reagents

1. Distilled water, carbon dioxide free.

2. Fisher Certified buffer solution, S - B - 98, pH 4.00 @ 25°C.

3. Fisher Certified buffer solution, S - B - 116, pH 10.00 @ 25°C.

b Apparatus

1. pH meter with glass-calomel electrode pair.

2 Filter paper.

c. Procedure Weigh approximately 10 grams (g) of sample into a beaker, add 25 milliliters (ml) of distilled water and stir for a minimum of 5 minutes. Filter the mixture and determine pH on the filtrate using a pH meter with glass-calomel electrode pair which has been calibrated against the known buffer prior to use.

d. Report the results of a minimum of 2 determinations and their average.

4.5.3 Assay. Assay shall be determined in accordance with the following:

a. Reagents:

1 Sodium chloride.

2. Sodium hydroxide.

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b. Apparatus: Cassia flask, 110 ml.

c. Preparation of sodium chloride solution: Add 300 g of sodium chloride to one liter of distilled water. Stir the solution and adjust the pH to between 10 and 11 using concentrated sodium hydroxide solution Stir the solution a minimum of 20 minutes or until essentially all the salt dissolves. Let the solution stand a minimum of 2 hours. Filter the supernatant liquid directly into a clean bottle.

d Procedure: Pipet 10 ml of the sample into a 110 ml Cassia volumetric flask containing 75 ml of distilled water at 21 ± 1°C and 5 ml of concentrated sodium chloride solution. Stopper the flask and invert 20 times to permit the trapped air to agitate (almost emulsify) the sample and water. Add sufficient 21 ± 1°C water to fill the graduated column to about 109.5 ml. Stopper the flask and invert 20 times to complete the mixing. Place the Cassia flask in a 21 ± 1°C water bath until the sample separates from the water and collects in the graduated column Determine the volume of the plasticizer in the graduated column by reading the bottom of the meniscus The oil layer must be clear; if not, make a new determination. (TABLE II provides a range of percent assay versus volume remaining in column in tabular form for information.)

e Calculation Assay, volume % = [0.10417(A) + 0.0550] (100)

where 0.10417 = density factor
A = volume of plasticizer remaining in column, ml
0.0550 = solubility constant

TABLE II. Volume of plasticizer remaining in column versus assay

Volume remaining in column, ml	Diethylene glycol dibutyl ether (percent)
4.0	47
5.0	58
6.0	68
7.0	78
8.0	89
9.0	99

f. Report the results of a minimum of 2 determinations and their average.

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4.5.4 Water content. The water content shall be determined in accordance with 4 5.4.1 or 4.5.4.2. In the event of controversy the results obtained by the method of 4 5.4.1 shall be determinative.

4 5.4.1 Preferred method for water content. The preferred method for determination of the weight percent (wt%) water content shall be in accordance with OD 18893.

4.5.4.2 Alternate method for water content. The alternate method for determination of water content shall be in accordance with the following:

a. Reagents.

1. Dry toluene (less than 0.005 percent water).

2. Methylene blue, very finely ground.

b. Apparatus:

1. Moisture trap, coated with Hydrophobic Monomers (see 6.4)

2 Erlenmeyer flask, 500 ml, 24/40 standard taper (ST)

3 "Glas-Col" mantle heating element, 500 ml Erlenmeyer

4 West Condenser, 24/40 ST (Corning 2740, or equal) with 300 ml jacket.

5 Rubber tubing 3/16 inch inside diameter (approximately 4 inches long).

c. Procedure: Add 24.9 to 25.1 g of sample, weighed to the nearest 0.01 g, to 100 g of dry toluene in an Erlenmeyer flask, stopper the flask, and mix well. Cover the flask with a "Glas-Col" mantle. Attach a piece of rubber tubing to the tip of the calibrated capillary of the water trap, and close the tubing Add approximately 2 ml of dry toluene to the moisture trap, and approximately 1 mg of finely ground methylene blue Assemble the apparatus and reflux the toluene with the sample for at least 1 hour at a rate of approximately 120 drops per minute Allow the apparatus to cool for approximately 15 minutes, then wash the condenser walls with three 5 to 10 ml portions of dry toluene Force the water column back and forth in the bottom of the moisture trap to collect all the water and to mix the methylene blue thoroughly with the water, then slowly open tubing to move the blue tinted water, as an unbroken column into the graduated capillary.

d Calculation Water content, wt% = (A)(0.04) x 100

where A = water reading, ml
0.04 = weight factor

e. Report the results of a minimum of 2 determinations and their average

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4.5.5 Color. Color shall be determined in accordance with the following:

a. Apparatus: Hellige varnish comparator, model number 605 equipped with number 620C-40 and number 620C-42 color disks, and sample tubes number 605V-T, or equivalent.

b. Procedure: Place the comparator so that the oval glass plate in the back of the instrument faces directly toward the light, taking care that pronounced shadows do not cause uneven illumination of the observation fields. Indirect natural lighting should be used. Insert a color disk, placing the center of the disk on the ring so that the number plates face the operator. Fill a clean, dry tube with a sufficient amount of sample to provide full exposure in the observation field. Place the tube in the right opening of the apparatus. Observe the sample and standard through the two central openings of the apparatus. Rotate the color disk until a color match is found between the sample and a standard. Record the number of the standard. If an exact color match cannot be obtained, determine the color number by interpolation of the standard numbers between which the sample lies.

c. Report the results of a minimum of 2 determinations.

4.5.6 Specific gravity. Specific gravity, 25/25°C, shall be determined in accordance with the following:

a. Apparatus

1. Fisher chain gravitometer or equivalent.

2. Thermometer, -20 to 100°C, with 1°C subdivisions

b. Procedure. Level the balance. With the roller weight at 0.0 and the movable scale at 0.000, place the empty cylinder on the platform so that the plummet will swing freely in the cylinder. Check the zero balance of the instrument. If necessary, adjust to zero. Mix a portion of the sample for a minimum of 1 minute and transfer an amount to the cylinder sufficient to cover the entire plummet. Place the cylinder containing the sample and a thermometer into a cooling bath until the cylinder and contents reach a temperature at a minimum of 1°C below that specified for the test. Remove the cylinder from the bath and remove the thermometer. Place the cylinder on the balance with the plummet inserted into the sample. When the temperature rises to 25 ± 1°C as indicated by the plummet thermometer, move the rider to the right, notch by notch, until the position is found which is just insufficient to cause the plummet to drop. This will give the first 2 units for the specific gravity of the liquid. Bring to a final balance by raising the chain weight slide. The second and third decimal places are read off the numbers on the graduated scale.

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c. Report the results of a minimum of 2 determinations and their average.

4.6 Packaging inspection The packaging, packing and marking shall be inspected to verify conformance with the requirements of Section 5.

5. PACKAGING

5.1 Packaging and packing Unless otherwise specified in the contract (see 6.2), packaging and packing shall be Level C.

5.1.1 Level C. Unless otherwise specified in the contract (see 6.2), packaging and packing of diethylene glycol dibutyl ether shall be in accordance with standard commerical practice applicable to the type of material The packaging and packing shall be of such construction and materials that the contents will be adequately protected against loss or contamination. Container size shall be as specified in the contract (see 6.2) Containers shall conform to Uniform Freight Classification, National Motor Freight Classification or to rules of other carriers applicable to the mode of transportation and shall be suitable for indoor storage at 32°C (90°F) maximum.

5.2 Marking. In addition to any special marking required by the contract (see 6.2), each container shall be marked in accordance with MIL-STD-129 and Code of Federal Regulations 49 CFR 171-179 Marking shall include, but not be limited to, the following information:

a. Title, number and date of this specification
b. Manufacturer's name and location
c Material trade name
d. Net weight
e Lot number, batch number(s), and date of manufacture
f. Storage conditions
g Contract or purchase order number
h. Shelf life (see 6.3)

6. NOTES

6.1 Intended use. Diethylene glycol dibutyl ether in accordance with this specification is intended for use as an ingredient in a solid propellant for the CCU-22/A impulse cartridge and other cartridge devices.

6.2 Ordering data Procurement documents should specify the following:

6.2.1 Procurement requirements.

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a. Title, number and date of this specification
b. Quantity required
c. Place of delivery
d. Inspection conditions when other than as specified (see 4.2)
e. Lot size if other than as specified (see 4.3.1)
f. Packaging requirements if other than as specified (see 5.1)
g. Size of container required (see 5.1)
h. Any special markings required (see 5.2)

6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraph herein:

Paragraph	Data Requirement	Applicable DID*
4.4	Quality conformance inspection data	-

*DID's (Data Item Description/DD Form 1664) for the above requirements will be documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DID's when specified on DD Form 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

6.3 Shelf life. The nominal shelf life of this material is 6 months for the date of manufacture when stored under ordinary warehouse conditions. Material that has exceeded this shelf life should be retested to the requirements of this specification prior to use.

6.4 Hydrophobic monomers. Desicote has been used satisfactorily in the water content test of 4.5.4 for coating the moisture trap.

6.5 Suggested source of supply. A product that has met the requirements of this specification in past procurement actions is TP-90B, ® manufactured by Thiokol Chemical Corporation. This information is for the convenience of the procuring activity and is not to be construed as a waiver of any requirement of this specification nor as any limitation of additional potential sources of supply.

6.6 Supersession information. MIL-D-82659 is intended to be used in lieu of AS 3058 (Code Ident 30003) dated 6 December 1974 for Naval Sea Systems Command procurement.

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Preparing Activity:
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Project Number:
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DOCUMENT IDENTIFIER AND TITLE
MIL-D-82659(OS), Diethylene Glycol Dibutyl Ether

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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